

Article

Segmented Bias Correction of ERA5 100 m Wind Speed for Wind-Resource Assessment in Complex Terrain

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Abstract

ERA5 100 m wind speed provides long-term and spatially continuous information for regional wind-resource assessment, but its grid-scale representation may introduce terrain- and wind-regime-dependent biases in complex terrain. This study evaluated ERA5 using 459,630 valid hourly observation–reanalysis pairs from 64 wind masts and developed a segmented residual-correction framework incorporating wind-regime, terrain, location, and temporal predictors. Three statistical correction methods and four tree-based models were compared under identical training and validation samples. The main analysis used a stratified random holdout, supplemented by chronological and tower-level spatial holdouts and repeated high-wind experiments. In the random validation subset, raw ERA5 yielded $R = 0.669$, $R^2 = 0.307$, $RMSE = 2.261 \text{ m s}^{-1}$, $MAE = 1.664 \text{ m s}^{-1}$, and $ME = -0.904 \text{ m s}^{-1}$. Random forest achieved the best paired hourly performance, increasing R to 0.850 and reducing $RMSE$ and MAE to 1.433 and 1.078 m s^{-1} , respectively. Its $RMSE$ reductions remained positive but decreased to 14.15% and 14.47% under chronological and spatial holdouts. Terrain relief was strongly associated with tower-level raw ERA5 errors, although its controlled inclusion produced only a modest additional $RMSE$ reduction of approximately 0.7%. Pooling the $>9 \text{ m s}^{-1}$ training tail improved high-wind stability and outperformed a separately trained $>12 \text{ m s}^{-1}$ model in 17 of 20 experiments. Random forest was preferable for paired hourly reconstruction, whereas quantile mapping more closely reproduced pooled theoretical wind-energy indicators. These results show that ERA5 bias correction should be selected according to the intended application and evaluated using complementary temporal, spatial, distributional, and high-wind diagnostics.

Keywords: ERA5; wind speed correction; wind-resource assessment; random forest; quantile mapping; wind power density; wind mast; complex terrain



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1. Introduction

Wind energy is a major component of the renewable-energy transition, and its continued expansion requires reliable information on wind resources at turbine-relevant heights. International energy assessments indicate that wind power will remain one of the principal contributors to renewable electricity growth, while climate-mitigation pathways require rapid expansion of low-carbon electricity supply [1,2]. In practical wind-energy development, site-level wind-resource assessment commonly relies on wind-mast measurements, measure–correlate–predict procedures, and turbine power-performance analysis. These procedures are designed to relate short-term local measurements to long-term wind conditions and to reduce uncertainty in annual energy production estimates [3,4]. However,

wind-mast observations are usually sparse, expensive to maintain, and spatially limited, especially in mountainous or hilly regions. Therefore, gridded meteorological datasets are frequently used to provide long-term background wind information, complement local observations, and support regional wind-resource mapping [5,6].

Wind-resource assessment therefore relies on complementary data sources. In-situ wind-mast measurements provide direct site-specific observations and support measure–correlate–predict procedures and standardized power-performance evaluation [3,4]. Gridded climatologies and wind-resource products complement sparse mast networks by providing long-term background conditions and spatial context [5,6]. Remote-sensing systems can add height-resolved wind information, while operational turbine or SCADA records, where available and quality-controlled, can link meteorological conditions with realized power performance; these sources complement rather than replace standardized mast-based measurements [3,4]. Numerical weather prediction and reanalysis products further provide spatially continuous and temporally consistent meteorological fields. In complex terrain, these data are commonly interpreted together with terrain information because local flow can vary at spatial scales smaller than those resolved by the atmospheric-model grid.

Among gridded products, ERA5 is particularly attractive for wind-resource applications because it provides globally consistent, long-term hourly meteorological fields, including eastward and northward wind components at 100 m [7,8]. Comparative studies have shown that ERA5 often performs favorably relative to older global reanalyses, although its accuracy depends on terrain, height, spatial aggregation, and the evaluation metric [9,10]. Bias-corrected reanalysis has been used to improve wind-power and resource estimates [11,12], while regional or higher-resolution reanalyses can add value where local terrain and meteorological structure are better resolved [13–15]. Recent studies have further demonstrated spatially varying wind corrections over China [16], application-dependent ERA5 wind-energy biases over the United States [17], benefits from local bias adjustment near the Moroccan Atlantic coast [18], height- and terrain-dependent reanalysis performance in lidar validation [19], and the potential of deep-learning correction for ERA5 hourly wind speed [20]. Taken together, these studies show both the practical advantages of ERA5 and the limitations imposed by grid-scale orography and unresolved sub-grid flow, thereby motivating height-matched local validation and bias correction before ERA5 is used for site-level wind-resource assessment.

Previous ERA5 wind-correction studies have used mean or distributional adjustments to reduce biases in reanalysis-based wind-resource estimates [11,12,17,18]. Quantile mapping has also been widely applied in climate-model bias correction [21–23] and is included here as a general distribution-matching benchmark rather than as a wind-specific paired prediction model. These statistical approaches are transparent and computationally efficient, but they do not explicitly represent nonlinear interactions among wind regime, season, location, and terrain. Moreover, improvement in the marginal distribution does not necessarily imply lower paired hourly errors. Recent work has extended ERA5 wind-speed correction to long short-term memory and Transformer architectures [20], demonstrating the potential value of temporal context when sequential predictors are available.

For the present problem, the predictors are heterogeneous tabular variables and the objective is paired residual correction rather than multi-step sequence forecasting. We therefore selected random forest [24] as a bagging reference and XGBoost [25,26], LightGBM [27], and CatBoost [28] as three boosting implementations with different regularization and tree-growing mechanisms. Comparing these models using identical predictors, samples, wind-speed regimes, and validation splits isolates the effect of the learning strategy. This design also tests whether greater model flexibility improves paired accuracy without as-

suming that the model with the lowest paired error will also be optimal for wind-speed distributions or wind-energy indicators.

This study develops a segmented residual-correction framework for ERA5 100 m wind speed using matched wind-mast observations in complex terrain. Its novelty lies in the joint treatment of five issues that are often evaluated separately: (i) comparison of three transparent statistical baselines and four tree ensembles using the same hourly samples; (ii) regime-dependent residual learning with training bins separated from evaluation bins; (iii) random, temporal-extrapolation, and spatial tower-holdout validation; (iv) explicit stability testing of the sparse high-wind tail through repeated stratified splitting and tower-block bootstrap; and (v) parallel assessment of paired errors, pooled wind-energy indicators, exceedance probability, and tower-level two-parameter Weibull characteristics. The remainder of the paper is organized as follows. Section 2 describes the wind-mast and ERA5 data, wind-component conversion, quality control, terrain-relief descriptor, segmented residual-learning framework, validation design, and theoretical wind-energy and Weibull diagnostics. Section 3 presents the raw ERA5 error characteristics, model comparison, wind-regime and mast-level performance, robustness, high-wind stability, terrain-complexity evidence and Weibull goodness-of-fit results. Section 4 discusses the physical and methodological implications and limitations, including the absence of coincident SCADA generation data. Section 5 summarizes the main conclusions.

2. Materials and Methods

2.1. Study Area and Data

The study used matched wind-mast observations and ERA5 100 m wind speed over a region containing plains, hills, and mountains. The network comprised 64 wind masts. Each mast contributed a fixed 365-day hourly archive window, giving 560,640 hourly archive slots in total. After unavailable or missing hourly slots were excluded, 459,630 valid 100 m wind-mast observation–ERA5 pairs remained. The available paired records spanned 9 January 2013 15:00 UTC to 28 February 2025 15:00 UTC, although the one-year archive window differed among masts. The main stratified random experiment used 367,704 records for training and 91,926 for validation. The tower network extended from 109.024 to 116.091° E and from 29.133 to 32.396° N. Across all valid paired records, the ERA5 wind-speed ranges contained 188,762 records at 0–3 m/s, 188,681 at 3–6 m/s, 69,492 at 6–9 m/s, 11,533 at 9–12 m/s, and 1162 above 12 m/s. Five ranges were retained for characterization and evaluation, whereas the final model pooled the 9–12 and >12 m/s ranges into a >9 m/s training tail to improve statistical stability. Shuttle Radar Topography Mission data were used as the elevation source [29].

ERA5 provides the eastward (u_{100}) and northward (v_{100}) wind components at 100 m. The scalar ERA5 100 m wind speed used for matching and subsequent analysis was calculated hourly from these two orthogonal components as

$$V_i = \sqrt{u_{100,i}^2 + v_{100,i}^2}$$

where $u_{100,i}$ and $v_{100,i}$ are the eastward and northward ERA5 wind components at 100 m, respectively, and V_i is the corresponding scalar wind speed. The pre-calculated scalar value was then matched to the 100 m AGL wind-mast observation at the same time.

Terrain complexity was represented by a 5-km terrain-relief index derived from the DEM. For each wind mast j , terrain relief was calculated as the difference between the maximum and minimum terrain elevations within a circular area of 5-km radius centered on the mast:

$$TR_{5,j} = z_{max,j}^{(5\text{ km})} - z_{min,j}^{(5\text{ km})}$$

where $TR_{5,j}$ is the terrain relief (m) for mast j ; $z_{\max,j}$ and $z_{\min,j}$ denote the maximum and minimum DEM elevations, respectively, within the 5-km-radius circular neighborhood. Larger terrain-relief values indicate greater local topographic variation.

Figure 1 shows the terrain background and the distribution of the 64 wind masts. The tower network covers both low-elevation areas and elevated terrain, which allows the correction framework to be evaluated under different topographic conditions. The terrain information was used to describe the study area and to construct auxiliary predictors.

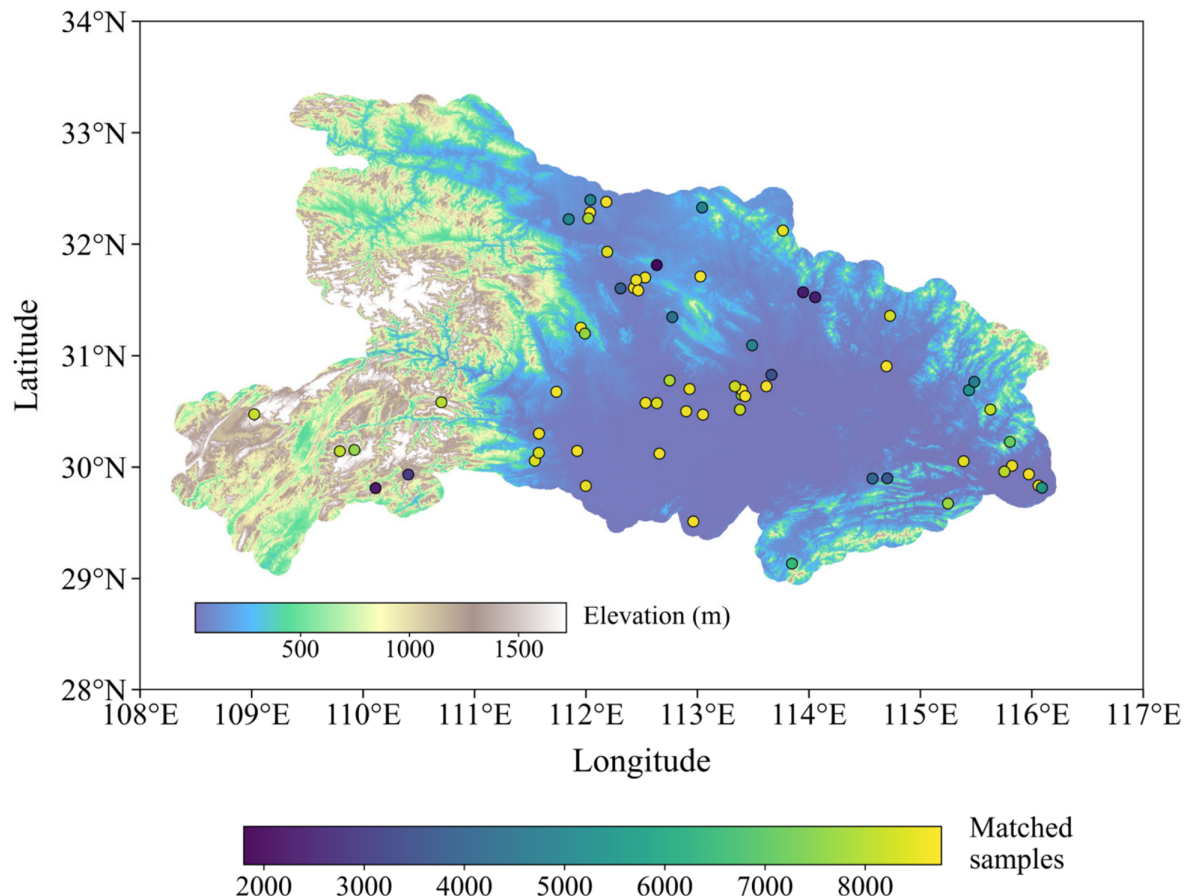


Figure 1. Study area, terrain elevation, and distribution of the 64 wind masts. Longitude and latitude are shown explicitly on the map axes. The background denotes ground elevation above mean sea level rather than mast height. All wind-speed observations used in the analysis were measured at 100 m AGL. Mast color denotes the number of valid matched observation-ERA5 samples.

The physical heights of the 64 wind masts are not identical; however, all selected masts extend above 100 m and provide wind-speed measurements at the common 100 m above-ground-level (AGL) measurement level. To ensure strict height consistency with the ERA5 100 m wind product, only the observations measured at 100 m AGL were extracted and used in the present analysis. Therefore, no vertical extrapolation of the wind-mast observations was required. The elevation shown in Figure 1 represents ground elevation above mean sea level rather than the physical height of the wind masts.

2.2. Data Preprocessing and Quality Control

The analysis began from the 560,640 hourly archive slots of the 64 mast-specific 365-day windows. Storage flags such as −9999 and hourly slots without a valid 100 m mast observation or matched ERA5 value were treated as unavailable rather than as observations, leaving 459,630 valid paired records. No additional valid pair was removed by the 0–40 m/s

physical-range screening. Wind-mast timestamps were interpreted in Beijing Time and converted to UTC ($\text{UTC} = \text{BJT} - 8 \text{ h}$) before ERA5 matching, and the nearest ERA5 grid point was used. Duplicate tower–time pairs were excluded before model splitting. The main 80:20 stratified split contained 367,704 training records and 91,926 validation records. The cleaned records were assigned to five evaluation bins (0–3, 3–6, 6–9, 9–12, and >12 m/s). For model training, the last two evaluation bins were pooled into a single >9 m/s tail to avoid fitting an unstable model to the sparse >12 m/s subset. This local-regime formulation follows the general statistical principle that non-stationary response relationships may be represented by separate models [30].

To visualize the data-level differences between ERA5 and the matched observations, Figure 2 compares their probability-density distributions within the five ERA5-defined wind-speed strata of the main training subset. The corresponding training-sample counts are 151,010, 150,945, 55,594, 9226, and 929 for 0–3, 3–6, 6–9, 9–12, and >12 m/s, respectively. Because the strata are defined according to raw ERA5 wind speed, the observed distributions are not constrained to the same intervals and are generally broader than the ERA5 distributions. The changing degree and direction of distributional mismatch across the five strata show that the ERA5 error cannot be represented adequately by a single constant offset and provide direct data-level support for the segmented residual-correction framework.

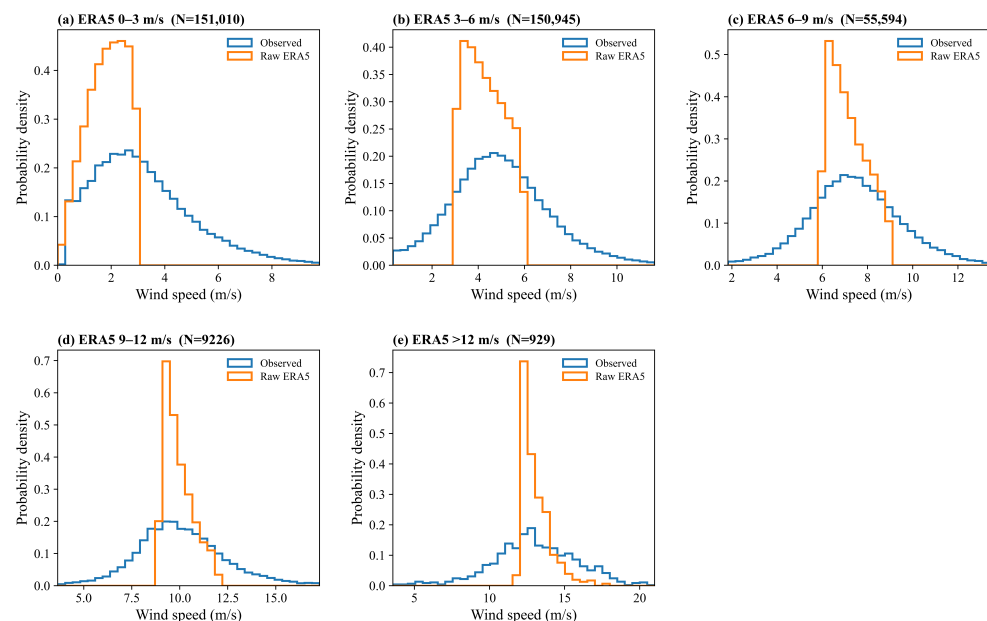


Figure 2. Probability-density distributions of raw ERA5 100 m wind speed and matched wind-mast observations at 100 m AGL within the five wind-speed strata of the main training subset. The strata are defined according to raw ERA5 wind speed. Five strata are retained for data characterization and evaluation, whereas the 9–12 and >12 m/s strata are pooled into a single >9 m/s regime during final model training to improve the statistical stability of the sparse high-wind tail.

Although five strata are retained for data characterization and evaluation, the final residual models use four training regimes (0–3, 3–6, 6–9, and >9 m/s). The 9–12 and >12 m/s strata are pooled only during model training to increase the effective sample size of the sparse high-wind tail, while their performance remains reported separately.

The primary predictor set included ERA5 wind speed; its square, cube, square-root, and $\log(1 + v)$ transformations; longitude, latitude, elevation, the 5-km terrain-relief index, longitude–latitude interaction, squared coordinates, and square-root elevation; and cyclic sine/cosine representations of hour, month, and day of year. The primary model did not use tower-history bias descriptors, so it can be applied at towers without prior observa-

tions. Three training-only descriptors (mean and median OBS-ERA5 residual and mean OBS/ERA5 ratio) were calculated only for a dedicated ablation experiment. All feature construction, quantile mapping, post-calibration, and parameter selection used training data only.

2.3. Segmented Residual-Learning Framework

The correction framework used residual learning. For sample i , let x_i denote ERA5 100 m wind speed, y_i the matched observation, X_i the predictor vector, and e_i the residual:

$$e_i = y_i - x_i \quad (1)$$

where e_i is the residual for sample i , y_i is the observed wind-mast wind speed at 100 m AGL, and x_i is the matched ERA5 100 m wind speed. For training wind-speed regime b , a regression model f_b was fitted to estimate the residual:

$$\hat{e}_i = f_b(X_i) \quad (2)$$

where $f_b(\cdot)$ denotes the regression model fitted for training regime b , X_i is the predictor vector for sample i , and $\hat{e}_i$ is the predicted ERA5 residual. The corrected wind speed was obtained by adding the predicted residual to ERA5:

$$\hat{y}_i = x_i + \hat{e}_i \quad (3)$$

where $\hat{y}_i$ is the corrected 100 m wind speed. Thus, the framework predicts the ERA5 error rather than the wind speed directly.

Separate residual models were trained for the 0–3, 3–6, 6–9, and >9 m/s regimes. Pooling the upper tail increased its training sample from 929 records above 12 m/s to 10,155 records above 9 m/s, while evaluation retained separate 9–12 and >12 m/s classes. This design reduced reliance on an unstable, data-sparse >12 m/s model without removing upper-tail diagnostics. As summarized in Figure 3, the workflow comprises data preparation and regime definition, segmented residual modeling with internal cross-validation, and random, temporal, spatial, repeated-split, and tower-block evaluations. Held-out predictions were subsequently assessed using paired wind-speed metrics and theoretical wind-energy diagnostics based on air density at 100 m AGL and a normalized turbine power curve.

2.4. Correction Methods and Model Configuration

To make the model comparison hypothesis-driven, we selected three transparent statistical baselines and four tree-ensemble models. Mean-bias correction and linear calibration represent first-order adjustments of the systematic error, whereas empirical quantile mapping [21–23] provides a distribution-oriented benchmark. Random forest [24] was selected as the representative bagging method because bootstrap aggregation and randomized feature selection generally provide stable nonlinear regression and reduce variance. XGBoost [25,26], LightGBM [27], and CatBoost [28] were selected as three representative boosting implementations because they sequentially reduce residual errors but differ in regularization, tree growth, sampling, and leaf construction. Tree ensembles are appropriate for the present predictors because the inputs are heterogeneous tabular variables and the relationships among ERA5 wind speed, terrain, location, season, and residual error are potentially nonlinear. All four tree models were trained using the same residual target, predictor set, samples, wind-speed regimes, and validation splits so that performance differences reflect the learning algorithms rather than inconsistent experimental settings.

Deep sequence models such as LSTM and Transformer [20] were not included because the present input design uses contemporaneous tabular predictors without a prescribed temporal sequence. A fair comparison with sequence-learning models would require a different input construction, including lagged predictor windows, sequence-length selection, and a separate hyperparameter-tuning framework. The present comparison was therefore restricted to statistical methods and tree ensembles that could be evaluated using identical tabular inputs and validation samples. This design provides a controlled comparison among distribution matching, bagging, and boosting instead of an open-ended benchmark of unrelated model classes. The main characteristics, strengths, and limitations of the seven correction methods are summarized in Table 1.

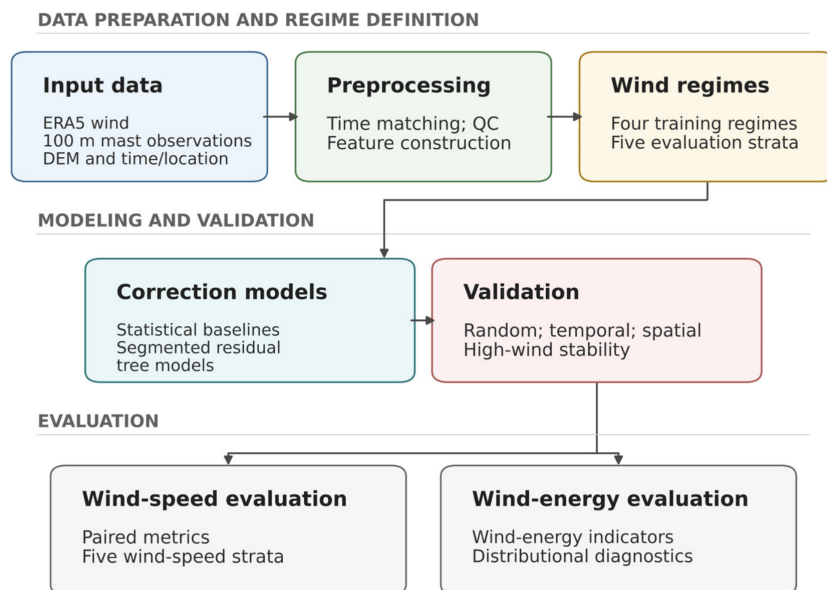


Figure 3. Methodological framework for ERA5 100 m wind-speed bias correction, robust validation, and wind-energy assessment. The workflow distinguishes five evaluation strata from four training regimes and routes held-out predictions to paired wind-speed evaluation and wind-energy diagnostics.

Table 1. Bias-correction methods considered in this study.

Method	Main Idea	Strengths	Limitations
Mean-bias correction	Add mean residual	Transparent; removes mean bias	No variance or regime correction
Linear calibration	Fit $OBS = aERA5 + b$	Simple linear adjustment	Weak for nonlinear terrain bias
Quantile mapping	Match empirical distributions	Good for wind-speed distribution	May not improve paired timing
Random forest	Average randomized trees	Robust nonlinear regression	May smooth extremes
XGBoost	Regularized boosting	High predictive capacity	Sensitive to tuning and imbalance
LightGBM	Histogram-based boosting	Efficient for large samples	May overfit without constraints
CatBoost	Ordered boosting	Reduces prediction shift	No guarantee of best skill

Mean-bias correction adds the mean training residual to each ERA5 value:

$$\hat{y}_i = x_i + \bar{e}_{train} \quad (4)$$

where $\bar{e}_{train}$ is the mean OBS–ERA5 residual calculated exclusively from the training sample. Linear calibration fits the observed wind speed as a linear function of ERA5:

$$\hat{y}_i = ax_i + b \quad (5)$$

where a and b are the slope and intercept estimated from the training observations only. Empirical quantile mapping transforms ERA5 through the training cumulative distributions:

$$\hat{y}_i = F_{y,train}^{-1}[F_{x,train}(x_i)] \quad (6)$$

where $F_{x,train}$ is the empirical cumulative distribution function (ECDF) of the training ERA5 wind speeds, and $F_{y,train}^{-1}$ is the inverse ECDF (quantile function) of the training observations. Quantile mapping is designed to reproduce a marginal distribution, whereas residual learning is designed to minimize paired prediction errors.

The tree-based models used the residual target defined in Equation (1). Hyperparameters were tuned separately for each algorithm and training wind-speed regime using three prespecified candidate configurations representing balanced, higher-complexity (“deeper”), and more strongly regularized model structures; the complete candidate settings are reported in Supplementary Table S2. All tuning was confined to the main training subset. When a wind-speed regime contained more than 50,000 training records, a reproducible random subset of 50,000 records was used for internal tuning with a fixed random seed of 42. Accordingly, 50,000 records were used for each of the 0–3, 3–6, and 6–9 m/s regimes, whereas all 10,155 records in the pooled >9 m/s regime were retained. Shuffled K-fold cross-validation used three folds for the first three regimes and five folds for the pooled >9 m/s regime. Within each fold, the model was fitted to the OBS–ERA5 residual, residual predictions were restricted to the 0.5th–99.5th percentiles of the fold-training residuals, and the resulting corrected wind speeds were restricted to 0–40 m/s. Candidate configurations were ranked by mean cross-validated RMSE, with RMSE standard deviation and candidate name used as secondary tie-breakers. The selected configuration was subsequently refitted using all available training records in the corresponding wind-speed regime. No record from the independent validation subset was used for hyperparameter selection.

After hyperparameter selection, an optional linear post-calibration was evaluated separately within each training regime. A model fitted to 75% of the regime-specific training records generated predictions for the remaining 25%, and a linear mapping from these predictions to observations was retained only when it did not increase RMSE and when its slope and intercept were within 0.75–1.25 and −3 to 3 m/s, respectively. This post-calibration step also used only the main training data.

The selected hyperparameter configurations and the complete candidate-level cross-validation results are provided in Supplementary Table S2. The selected configuration for each algorithm–regime combination is identified in the “Selected” column and highlighted in bold. All hyperparameter selection and optional post-calibration were performed exclusively within the main training subset.

All analyses and figures were implemented in Python 3.12.10 using NumPy 2.4.6, pandas 3.0.5, SciPy 1.18.0, scikit-learn 1.9.0, XGBoost 3.3.0, LightGBM 4.7.0, CatBoost 1.2.10, and Matplotlib 3.11.1. Cartographic processing used Cartopy 0.25.0, GeoPandas 1.1.4, and Rasterio 1.5.0. A fixed random seed was used for the main split, cross-validation, model fitting, repeated high-wind experiments, and bootstrap resampling. The same main

training and validation samples were used for all methods. Earlier records were used to train the temporal-extrapolation model, and groups of complete wind masts were withheld in the spatial experiment. No validation observations were used to fit residual models, empirical quantiles, post-calibration coefficients, or tower-history descriptors.

2.5. Validation Strategy and Evaluation Metrics

The main validation strategy used an 80:20 split stratified by the five ERA5 evaluation bins, yielding 91,926 validation records, including 233 records above 12 m/s. Two stricter experiments assessed transferability. Temporal extrapolation was trained on earlier records and evaluated on later records, while spatial tower holdout excluded complete wind-mast groups from training. Ablation experiments compared the primary segmented random forest with (i) an otherwise identical model without the 5-km terrain-relief feature, (ii) a tower-bias variant, and (iii) a global, non-segmented random forest residual model.

Six paired metrics were used to evaluate wind-speed correction: Pearson correlation coefficient (R), coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), normalized MAE (nMAE), and mean error (ME). R^2 was calculated directly as $1 - \text{SSE}/\text{SST}$ rather than as the square of R , and nMAE was normalized by the mean observed wind speed. R measures linear co-variation, whereas R^2 measures the fraction of observed variance explained after accounting for bias and amplitude error.

$$R = \frac{\sum_{i=1}^N (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2}} \quad (7)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (8)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (9)$$

$$\text{ME} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i) \quad (10)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (11)$$

$$\text{nMAE} = 100 \times \frac{\text{MAE}}{\bar{y}} \quad (12)$$

where N is the number of validation samples; y_i and $\hat{y}_i$ are the observed and predicted wind speeds, respectively; $\bar{y}$ and $\bar{\hat{y}}$ are their sample means. Equation (7) is the sample Pearson correlation coefficient; Equation (11) calculates R^2 directly from the residual sum of squares rather than as the square of R . RMSE emphasizes larger deviations, MAE measures the mean absolute deviation, nMAE expresses MAE relative to the mean observed wind speed, and ME indicates the signed systematic bias.

In this study, nMAE is a wind-speed error metric rather than a wind-power forecasting metric. In power-forecasting applications, MAE is often normalized by installed or rated power capacity because both the prediction target and the normalization quantity are expressed in power units. Here, however, the prediction target is wind speed (m/s), and no turbine-specific installed capacity is associated with each hourly wind-speed pair. Normalizing a wind-speed MAE by power capacity would therefore be dimensionally inconsistent. We consequently normalize MAE by the mean observed wind speed of the corresponding evaluation sample, producing a dimensionless relative wind-speed error. Wind-energy performance is assessed separately using WPD, capacity factor, equivalent full-load hours, exceedance probability, and Weibull diagnostics.

RMSE and MAE were retained because they weight large errors differently [31], while nMAE permits comparison relative to the mean wind climate. Improvement rates were calculated relative to raw ERA5. The best tree model was selected lexicographically: minimum RMSE, then minimum MAE if RMSE was tied, and finally minimum |ME|. R , R^2 , and nMAE were reported as complementary diagnostics rather than used to override that error-based criterion.

The six metrics were selected as complementary diagnostics because no single statistic fully characterizes paired wind-speed correction. Pearson's correlation coefficient (R) quantifies temporal co-variation between predictions and observations but is insensitive to systematic offset. The coefficient of determination (R^2), calculated directly from the residual sum of squares, measures the fraction of observed variance reproduced by the predictions. RMSE gives greater weight to large errors and is therefore sensitive to high-amplitude deviations that can strongly affect wind-energy calculations. MAE provides a more robust measure of the typical absolute error and is less dominated by a small number of extreme deviations. The mean-normalized MAE (nMAE) expresses MAE relative to the characteristic wind-speed scale of the evaluated sample, facilitating comparison among validation strategies with different mean wind climates. Finally, ME preserves the sign of the error and identifies systematic over- or underestimation. Together, these metrics describe association, explained variance, large-error sensitivity, typical error magnitude, relative error magnitude, and systematic bias.

No coincident, quality-controlled wind-farm SCADA or actual generation time series was available for the 64 wind-mast sites. Therefore, the study does not perform direct validation against realized wind-farm generation. The WPD, capacity factor, and equivalent full-load hours reported below are resource-derived theoretical diagnostics calculated from observed, raw, or corrected wind speed using a common air-density treatment and a generic normalized turbine power curve. They are used only to compare the consequences of different wind-speed correction methods and should not be interpreted as observed plant production or turbine-specific bankable energy yield.

Wind-energy impacts were evaluated using wind power density (WPD), capacity factor, and equivalent full-load hours. Air density was evaluated at 100 m above ground level. Because matched pressure and temperature were unavailable, density was estimated from the International Standard Atmosphere at an absolute altitude equal to terrain elevation plus 100 m. The mean validation air density was 1.175 kg m^{-3} . The standard-atmosphere treatment accounts for terrain elevation and the common 100 m AGL but not hourly thermodynamic variability.

$$z_i = h_{\text{terrain},i} + 100 \text{ m} \quad (13)$$

$$T_i = T_0 - Lz_i \quad (14)$$

$$p_i = p_0 \left(\frac{T_i}{T_0} \right)^{\frac{g}{R_d L}} \quad (15)$$

$$\rho_i = \frac{p_i}{R_d T_i} \quad (16)$$

$$\text{WPD}_i = \frac{1}{2} \rho_i v_i^3 \quad (17)$$

where z_i is the absolute altitude used for the standard-atmosphere calculation, $h_{\text{terrain},i}$ is terrain elevation above mean sea level, T_i is standard-atmosphere temperature, p_i is atmospheric pressure, ρ_i is air density, and v_i is wind speed for sample i . The constants are $T_0 = 288.15 \text{ K}$, $p_0 = 101,325 \text{ Pa}$, $L = 0.0065 \text{ K m}^{-1}$, $g = 9.80665 \text{ m s}^{-2}$, and $R_d = 287.05 \text{ J kg}^{-1} \text{ K}^{-1}$. Wind power density is expressed in W m^{-2} . This standard-

atmosphere treatment accounts for terrain elevation and the common 100 m AGL, but not hourly thermodynamic variability.

A normalized turbine power curve used cut-in, rated, and cut-out wind speeds of 3, 12, and 25 m/s, respectively [32,33]. Because the curve is generic and normalized, the resulting capacity factors are intended for method comparison rather than turbine-specific bankable energy estimates.

$$P^*(v) = \begin{cases} 0, & v < 3, \\ \frac{v^3 - 3^3}{12^3 - 3^3}, & 3 \leq v < 12, \\ 1, & 12 \leq v \leq 25, \\ 0, & v > 25. \end{cases} \quad (18)$$

where $P^*(v)$ is the normalized turbine power, and 3, 12, and 25 m/s represent the cut-in, rated, and cut-out wind speeds, respectively. The curve is generic and is used for inter-method comparison rather than turbine-specific bankable energy estimation.

$$CF = \frac{1}{N} \sum_{i=1}^N P^*(v_i) \quad (19)$$

where CF is the dimensionless mean normalized turbine output over the evaluated samples.

$$EFLH = 8760 \times CF \quad (20)$$

where EFLH is the equivalent full-load hours, and 8760 is the number of hours in a non-leap year. To examine wind-speed distributions in more detail, a two-parameter Weibull distribution was fitted separately at each wind mast with the location parameter fixed at zero. At least 50 positive observations were required. The shape parameter k , scale parameter c , Kolmogorov–Smirnov statistic, Weibull-derived WPD, and Weibull-derived capacity factor were calculated. A single Weibull fit was not applied to the pooled multi-tower sample because a mixture of local wind climates need not follow one Weibull distribution.

$$f(v) = \frac{k}{c} \left(\frac{v}{c} \right)^{k-1} \exp \left[- \left(\frac{v}{c} \right)^k \right], \quad v \geq 0 \quad (21)$$

where k is the dimensionless Weibull shape parameter, c is the Weibull scale parameter (m/s), and the location parameter is fixed at zero.

$$WPD_{Wei} = \frac{1}{2} \rho c^3 \Gamma \left(1 + \frac{3}{k} \right) \quad (22)$$

where $\Gamma(\cdot)$ is the gamma function, and ρ is the tower-specific air density estimated from terrain elevation plus 100 m AGL using the same standard-atmosphere method as in Equations (13)–(16).

$$CF_{Wei} = \int_0^\infty P^*(v) f(v) dv \quad (23)$$

where $P^*(v)$ is the normalized power curve defined in Equation (18). For each mast, the Kolmogorov–Smirnov statistic was calculated as

$$D = \sup_v | F_n(v) - F_W(v; k, c) | \quad (24)$$

A tower was counted as passing the Weibull goodness-of-fit check when the one-sample KS p -value was ≥ 0.05 . Because k and c were estimated from the same tower sample

used in the KS calculation, these p -values are interpreted as descriptive goodness-of-fit diagnostics rather than exact inferential tests.

High-wind stability was assessed in two ways. First, 20 repeated stratified train–test experiments compared raw ERA5 with an RF trained only on the >12 m/s segment and an RF trained on the pooled >9 m/s tail. In each repeat, the separate >12 m/s strategy used 929 training records and 233 test records, whereas the pooled >9 m/s strategy used 10,155 training records while retaining the same 233 >12 m/s test records. Mean, standard deviation, percentile-based 95% intervals, and the frequency of the lowest RMSE were recorded together with high-wind WPD bias. Second, a 1000-repetition tower-block bootstrap resampled complete high-wind validation towers to quantify sensitivity to mast composition.

3. Results

3.1. Error Characteristics of Raw ERA5 Wind Speed

Raw ERA5 100 m wind speed showed substantial error relative to the mast observations. In the stratified random validation sample ($N = 91,926$), R was 0.669, R^2 was 0.307, RMSE was 2.261 m/s, MAE was 1.664 m/s, nMAE was 34.66%, and ME was -0.904 m/s. Thus, ERA5 reproduced a substantial fraction of the temporal co-variation but retained a pronounced negative mean bias and considerable amplitude error.

Figure 4 illustrates the error structure. The scatter-density distribution is broad and lies predominantly below the 1:1 line, and the ERA5-minus-observed error distribution is centered near -0.90 m/s. Error magnitude varies markedly with wind-speed regime, and the highest ERA5 bin has the largest RMSE and MAE. These regime-dependent changes support a segmented correction rather than a single constant adjustment.

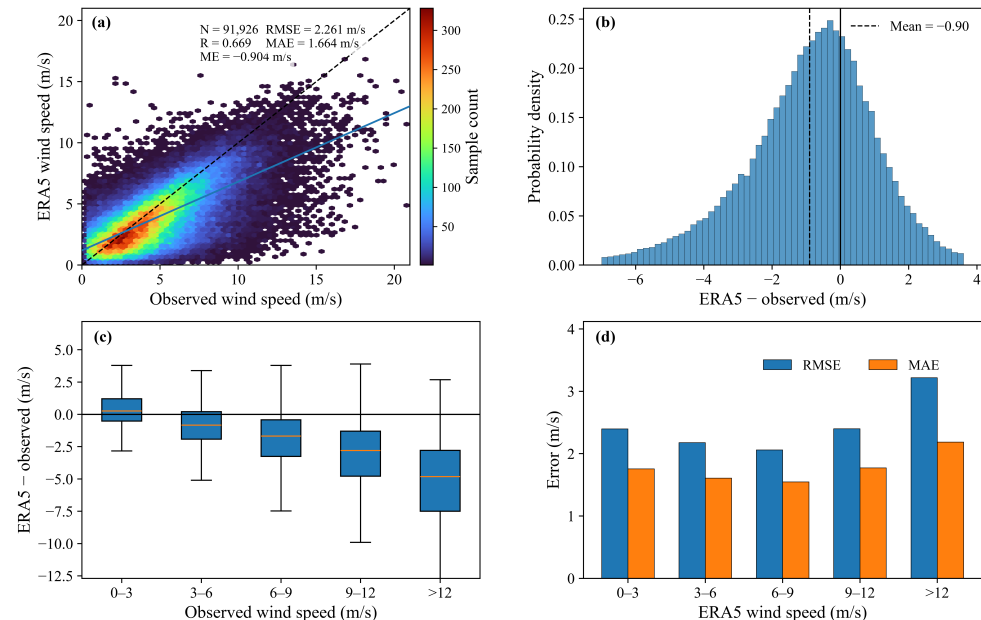


Figure 4. Error Error characteristics of raw ERA5 100 m wind speed. (a) Scatter-density relationship, in which the black dashed line denotes the 1:1 reference and the solid blue line denotes the fitted linear regression; (b) ERA5-minus-observed error distribution, in which the solid black vertical line denotes zero error and the black dashed vertical line denotes the mean error; (c) error by observed wind-speed range; and (d) RMSE and MAE by ERA5 wind-speed range.

3.2. Overall Performance of Bias-Correction Methods

Internal cross-validation selected the higher-complexity (“deeper”) candidate for all 16 algorithm–regime combinations. Across the four training wind-speed regimes, the

selected mean cross-validated RMSE ranged from 1.451 to 1.696 m/s for random forest, from 1.541 to 1.716 m/s for XGBoost, from 1.528 to 1.709 m/s for LightGBM, and from 1.647 to 1.752 m/s for CatBoost. These values are internal model-selection scores and should not be interpreted as the independent validation metrics. Because only three prespecified configurations were evaluated for each algorithm, the selected configuration represents the best candidate within the tested search set rather than a global hyperparameter optimum. Complete candidate-level results are reported in Supplementary Table S2.

Table 2 summarizes the stratified random-split results. Mean-bias correction reduced RMSE by 8.33%, and linear calibration reduced it by 10.66%. Quantile mapping nearly removed the mean bias but reduced paired RMSE by only 1.79% and slightly increased MAE by 0.89%, confirming that marginal-distribution adjustment does not necessarily improve sample-by-sample timing.

Among the segmented tree models, random forest gave the best paired hourly performance according to the prespecified $\text{RMSE} \rightarrow \text{MAE} \rightarrow |\text{ME}|$ rule. It increased R from 0.669 to 0.850 and R^2 from 0.307 to 0.722, reduced RMSE from 2.261 to 1.433 m/s, reduced MAE from 1.664 to 1.078 m/s, decreased nMAE from 34.66% to 22.45%, and reduced ME from -0.904 to 0.006 m/s. The RMSE and MAE improvements were 36.61% and 35.22%, respectively. The global RF residual model was slightly weaker ($\text{RMSE} = 1.455$ m/s), while LightGBM, XGBoost, and CatBoost yielded RMSE values of 1.579, 1.618, and 1.672 m/s, respectively.

Table 2. Overall validation metrics for raw ERA5 and correction methods under the stratified random split.

Method	R	R^2	RMSE (m/s)	MAE (m/s)	nMAE (%)	ME (m/s)	RMSE Imp. (%)	MAE Imp. (%)
Raw ERA5	0.669	0.307	2.261	1.664	34.66	-0.904	—	—
Mean-bias correction	0.669	0.418	2.073	1.560	32.49	0.003	8.33	6.26
Linear calibration	0.669	0.447	2.020	1.533	31.93	0.003	10.66	7.87
Quantile mapping	0.667	0.332	2.221	1.679	34.97	0.003	1.79	-0.89
Global RF residual	0.845	0.713	1.455	1.100	22.92	0.009	35.64	33.88
XGBoost	0.803	0.645	1.618	1.231	25.66	0.010	28.43	25.98
Random forest	0.850	0.722	1.433	1.078	22.45	0.006	36.61	35.22
LightGBM	0.814	0.662	1.579	1.199	24.98	0.009	30.19	27.92
CatBoost	0.788	0.621	1.672	1.272	26.50	0.012	26.07	23.55

Figure 5 compares the four segmented tree models. Random forest had the lowest RMSE and MAE, the highest R and R^2 , and the largest RMSE improvement. The consistency of these metrics, rather than a single isolated score, supports its selection as the primary paired hourly model.

3.3. Performance Across Wind-Speed Regimes and Wind Masts

Figure 6 shows performance by wind-speed range and wind mast. Random forest achieved the lowest RMSE among the four segmented tree models in all five ERA5 evaluation bins. Its RMSE values were 1.448, 1.453, 1.341, 1.400, and 1.482 m/s for 0–3, 3–6, 6–9, 9–12, and >12 m/s, corresponding to RMSE reductions of 39.57%, 33.21%, 34.90%, 41.65%, and 53.93% relative to raw ERA5. The >12 m/s point estimate is based on 233 validation

records and is therefore interpreted together with the repeated high-wind experiment in Section 3.8.

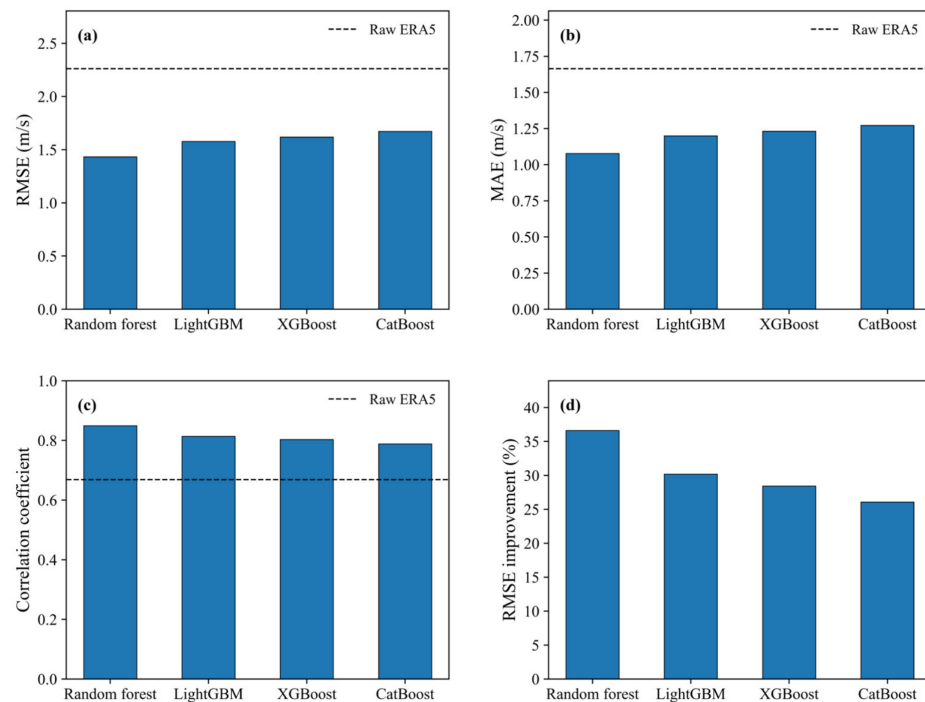


Figure 5. Overall comparison of four segmented tree-based correction models. (a) RMSE; (b) MAE; (c) Pearson correlation coefficient (R); and (d) RMSE improvement relative to raw ERA5. Black dashed lines in panels (a–c) denote the corresponding raw ERA5 reference levels.

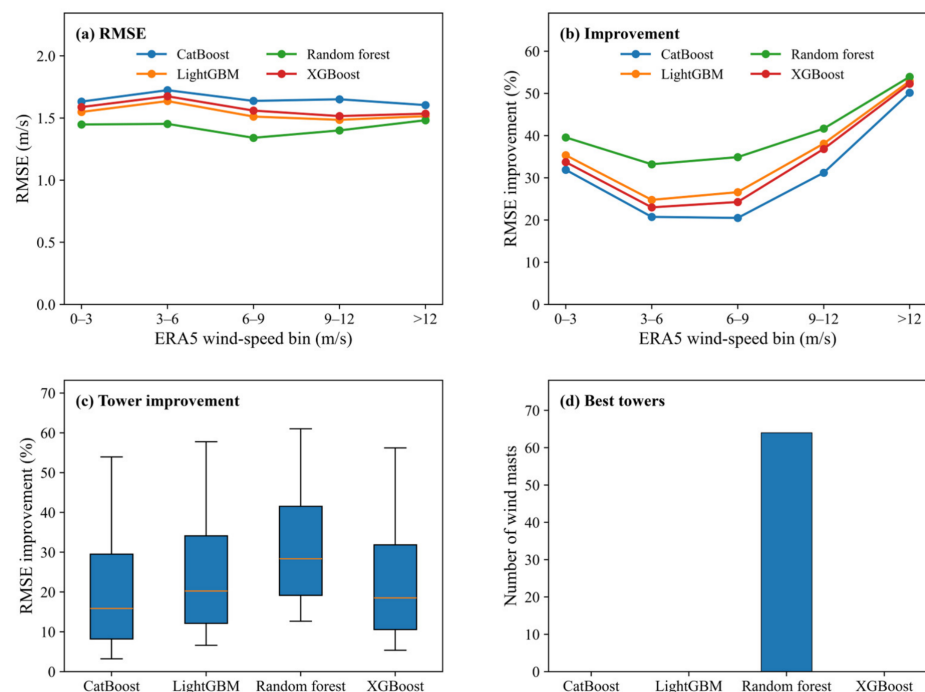


Figure 6. Performance of the four segmented tree-based correction models across wind-speed ranges and wind masts. (a) RMSE by wind-speed range; (b) RMSE improvement relative to raw ERA5 by wind-speed range; (c) tower-level RMSE improvement; and (d) number of towers showing the best performance for each model. In panels (a,b), the orange line represents LightGBM, as indicated in the legend; the orange horizontal lines in panel (c) denote the medians.

All 64 masts contained sufficient validation records for tower-level evaluation. The primary random forest reduced RMSE and MAE and increased both R and R^2 at every mast. Tower-level RMSE improvement ranged from 12.64% to 61.02%, with a median of 28.34%. When the four segmented tree models were ranked separately at each mast, random forest achieved the lowest RMSE at all 64 masts (Figure 6d). Detailed mast-specific statistics for all 64 sites, including raw ERA5 and corrected random forest performance, are provided in Supplementary Table S1.

3.4. Detailed Analysis of the Random Forest Correction

Figure 7 compares scatter-density distributions before and after random forest correction using the same 91,926 validation records. After correction, R increased from 0.669 to 0.850 and R^2 from 0.307 to 0.722, while RMSE decreased from 2.261 to 1.433 m/s, MAE from 1.664 to 1.078 m/s, nMAE from 34.66% to 22.45%, and ME from -0.904 to 0.006 m/s. The corrected density is substantially closer to the 1:1 relationship, showing that the improvement reflects both stronger paired agreement and lower error magnitude rather than a simple mean shift.

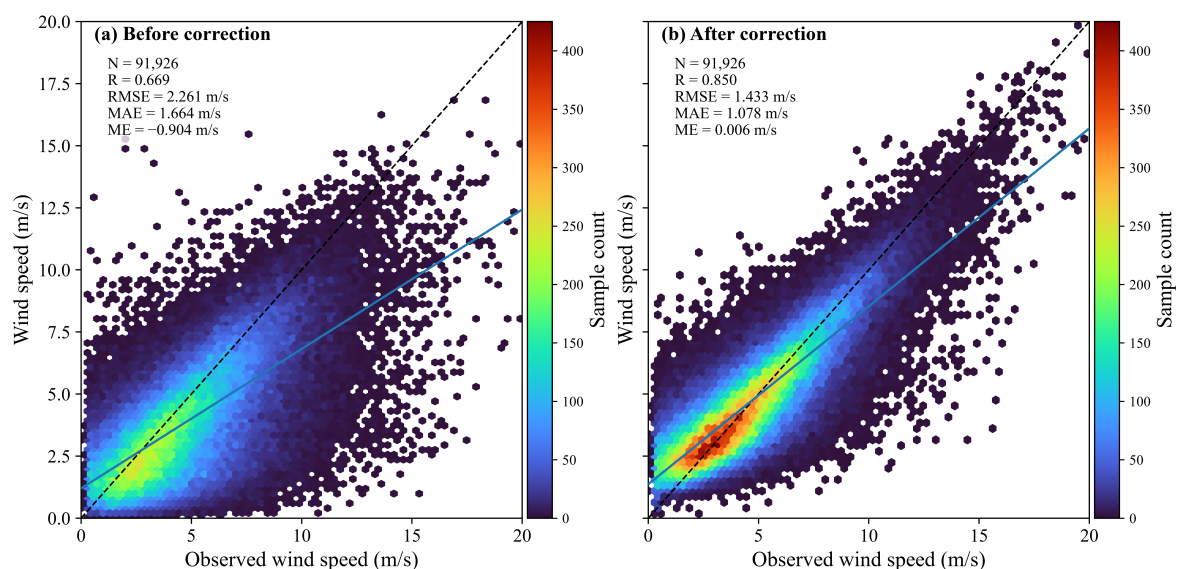


Figure 7. Scatter-Scatter-density comparison before and after random forest correction using the same random validation sample: (a) raw ERA5 wind speed and (b) random-forest-corrected wind speed. In both panels, the black dashed line denotes the 1:1 reference and the solid blue line denotes the fitted linear regression.

Figure 8 shows the temporal and wind-regime behavior of the primary random forest. Corrected diurnal and monthly means closely follow the observations, whereas raw ERA5 remains persistently low. The hourly corrected ME is generally near zero. Bias is also reduced in every observed-wind-speed class, although a negative median remains above 12 m/s. This residual upper-tail error explains why energy indicators are not completely restored.

3.5. Impacts on Wind-Energy Assessment

Table 3 summarizes resource-derived theoretical wind-energy indicators calculated using the common 100 m AGL air-density treatment and generic normalized turbine power curve. The observed-wind reference had a mean wind speed of 4.800 m/s, WPD of 135.74 W/m², capacity factor of 0.1167, and equivalent full-load hours of 1022.1 h. Raw ERA5 had a mean wind speed of 3.897 m/s and underestimated WPD by 42.99%, capacity factor by 45.16%, and equivalent full-load hours by 461.6 h.

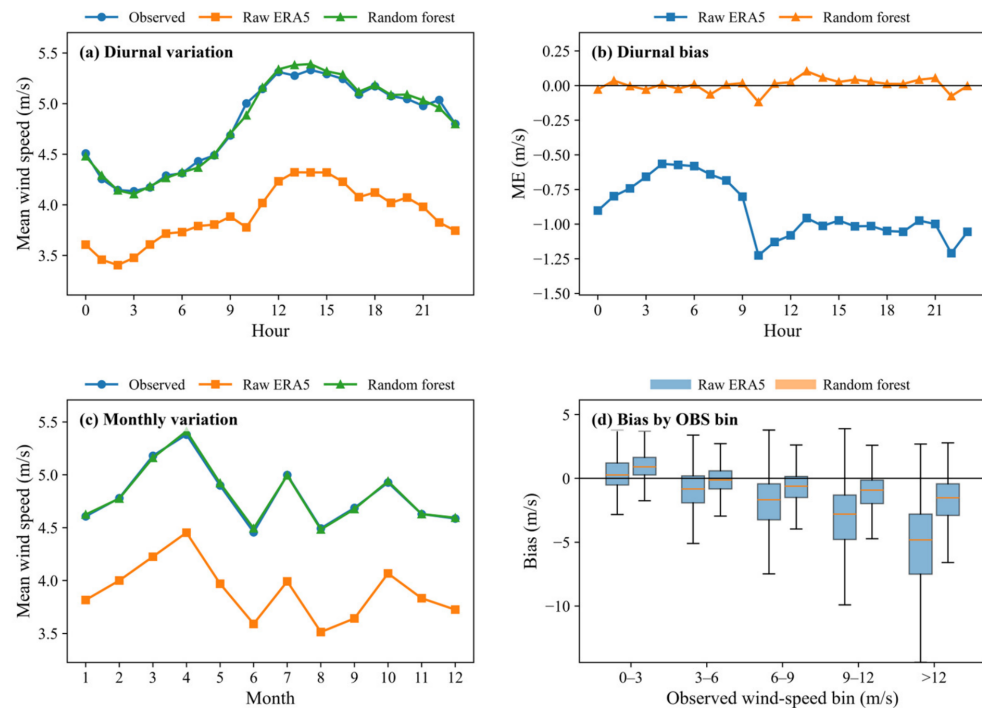


Figure 8. Temporal variation and wind-regime bias of random forest correction. (a) Diurnal mean wind speed; (b) diurnal ME; (c) monthly mean wind speed; (d) bias by observed wind-speed range.

Table 3. Resource-derived theoretical wind-energy indicators calculated using air density at terrain elevation plus 100 m AGL (mean validation density = 1.175 kg/m^3) and a generic normalized turbine power curve. These values are not observed wind-farm SCADA generation.

Dataset	Mean WS (m/s)	WPD (W/m^2)	WPD Bias (%)	CF	CF Bias (%)	EFLH (h)	EFLH Bias (h)
Observed	4.800	135.74	0.00	0.1167	0.00	1022.1	0.0
Raw ERA5	3.897	77.39	−42.99	0.0640	−45.16	560.5	−461.6
Mean-bias correction	4.803	116.50	−14.18	0.0992	−14.94	869.4	−152.8
Linear calibration	4.803	97.09	−28.47	0.0807	−30.80	707.3	−314.8
Quantile mapping	4.803	139.39	+2.69	0.1166	−0.10	1021.1	−1.0
Random forest	4.807	116.42	−14.23	0.0992	−14.98	869.0	−153.1

Random forest restored the paired mean wind speed to 4.807 m/s and improved all three energy indicators, but its WPD remained 14.23% below the observed-wind reference, consistent with residual smoothing of the upper tail. Quantile mapping behaved differently: although it was much weaker for paired hourly RMSE, its pooled WPD bias was +2.69%, capacity-factor bias was −0.10%, and equivalent-full-load-hour bias was −1.0 h. Thus, random forest is preferred for paired hourly reconstruction, whereas pooled quantile mapping more closely reproduces aggregate marginal-distribution energy diagnostics. These quantities are theoretical resource indicators and are not a validation against observed wind-farm SCADA generation.

Figure 9 visually compares mean wind speed, WPD, capacity factor, and equivalent full-load hours. Raw ERA5 is markedly low for all indicators. Random forest improves all four but retains tail-related energy underestimation; quantile mapping is closest to the pooled observed energy indicators.

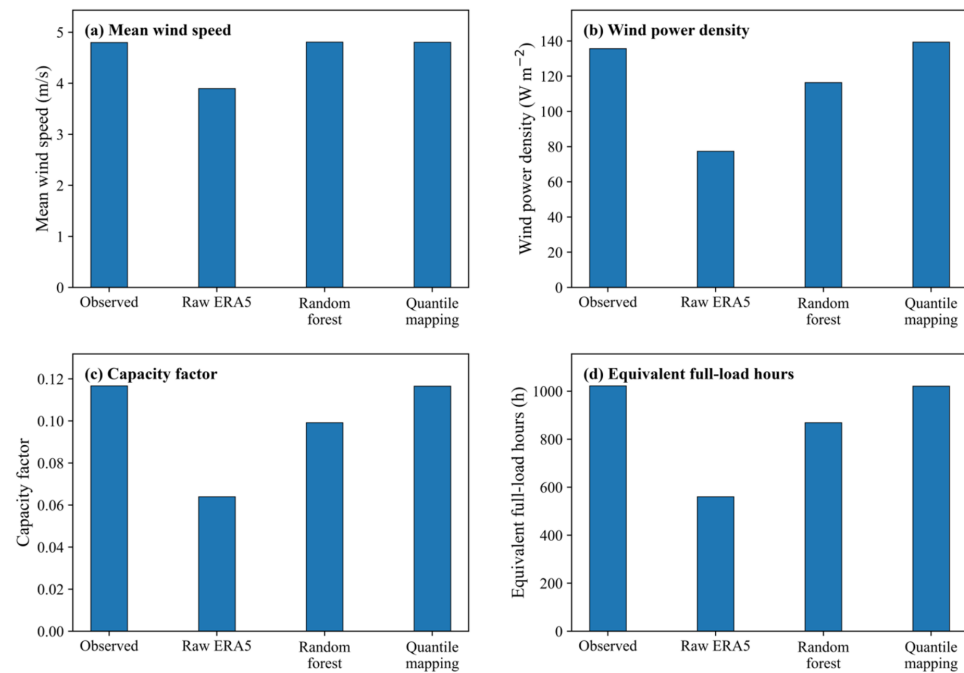


Figure 9. Effects of correction methods on mean wind speed, WPD, capacity factor, and equivalent full-load hours. Wind-energy quantities are resource-derived theoretical indicators based on a generic normalized turbine curve rather than observed wind-farm SCADA generation.

3.6. Robustness Under Temporal and Spatial Validation

Table 4 evaluates robustness and ablation. Under the main random split, the primary segmented random forest with the 5-km terrain-relief feature achieved $\text{RMSE} = 1.433 \text{ m/s}$. Removing the terrain-relief feature increased RMSE to 1.444 m/s, adding tower-history bias descriptors yielded $\text{RMSE} = 1.449 \text{ m/s}$, and the global non-segmented RF residual model yielded $\text{RMSE} = 1.455 \text{ m/s}$. Thus, the primary model remained the strongest of the tested RF variants and did not require tower-history descriptors.

Table 4. Robust validation and ablation results for the random forest correction.

Validation/Model	N	R	R^2	RMSE	MAE	nMAE (%)	ME	RMSE Imp. (%)
Random—Raw ERA5	91,926	0.669	0.307	2.261	1.664	34.66	−0.904	—
Random—RF primary	91,926	0.850	0.722	1.433	1.078	22.45	0.006	36.61
Random—RF without terrain relief	91,926	0.847	0.718	1.444	1.087	22.64	0.006	36.16
Random—RF + tower bias	91,926	0.846	0.716	1.449	1.092	22.75	0.007	35.93
Random—Global RF	91,926	0.845	0.713	1.455	1.100	22.92	0.009	35.64
Temporal—Raw ERA5	91,926	0.666	0.279	2.173	1.585	33.74	−0.839	—
Temporal—RF primary	91,926	0.689	0.468	1.866	1.414	30.09	−0.120	14.15
Spatial—Raw ERA5	98,972	0.677	0.352	2.268	1.650	34.96	−0.849	—
Spatial—RF primary	98,972	0.726	0.526	1.940	1.464	31.02	−0.001	14.47

The stricter transferability experiments produced smaller but still positive improvements. In temporal extrapolation, RMSE decreased from 2.173 to 1.866 m/s (14.15%), MAE from 1.585 to 1.414 m/s, R increased from 0.666 to 0.689, and R^2 from 0.279 to 0.468. In spatial tower holdout, RMSE decreased from 2.268 to 1.940 m/s (14.47%), MAE from 1.650 to 1.464 m/s, R increased from 0.677 to 0.726, and R^2 from 0.352 to 0.526. These reductions are smaller than the 36.61% improvement under random splitting and therefore indicate moderate rather than unrestricted temporal or spatial transferability.

Terrain complexity was strongly associated with the raw ERA5 error across the 64 masts. The 5-km terrain-relief index had a Pearson correlation of $r = 0.872$ with tower-level raw RMSE ($p < 0.001$) and a Spearman correlation of $\rho = 0.800$ ($p < 0.001$). Relief was also positively associated with RF RMSE improvement (Pearson $r = 0.754$; Spearman $\rho = 0.653$; both $p < 0.001$). In the controlled feature ablation, including terrain relief reduced RF RMSE from 1.444 to 1.433 m/s, an additional gain of 0.71%, and reduced MAE from 1.087 to 1.078 m/s (0.81%). The association is interpreted as co-variation across masts rather than as proof of a causal terrain mechanism.

Figure 10 illustrates the same contrast. Random splitting yields the largest improvement, whereas temporal and spatial transfer yield smaller gains. The spatial result showed a larger improvement in R and R^2 than the temporal result, while both validation strategies reduced RMSE and MAE and increased R and R^2 . The more moderate gains relative to the random split further highlight the importance of evaluating model transferability under multiple validation designs.

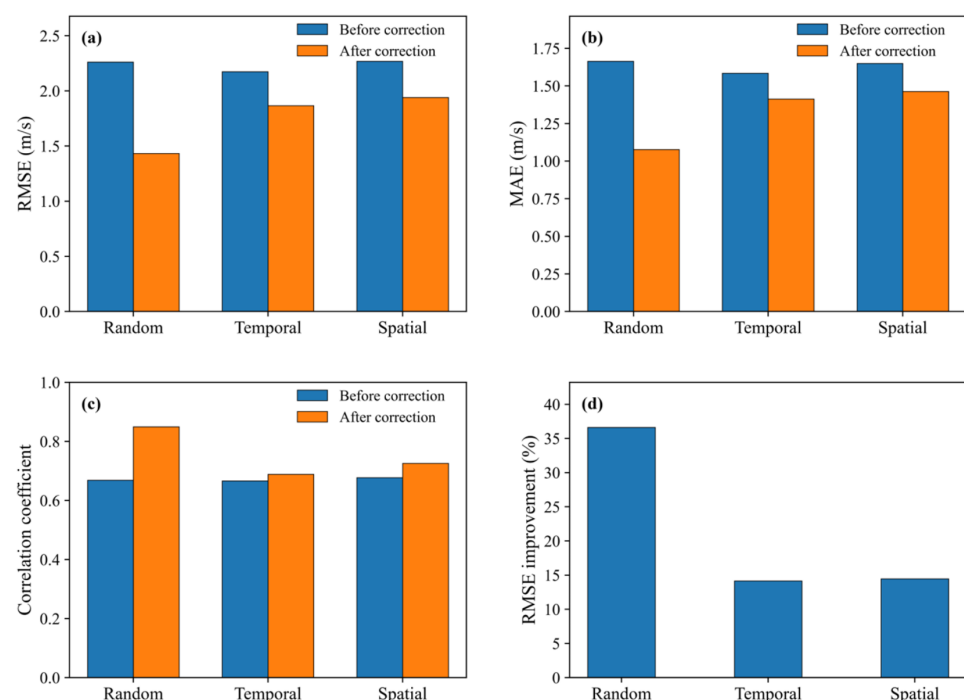


Figure 10. Random forest performance under random split, temporal extrapolation, and spatial tower holdout. Before correction denotes raw ERA5, whereas after correction denotes the primary pooled-tail random forest. (a) RMSE; (b) MAE; (c) Pearson correlation coefficient (R); and (d) RMSE improvement relative to raw ERA5.

3.7. Wind-Speed Exceedance Probability

Figure 11 explains the different energy outcomes. Raw ERA5 increasingly underestimates exceedance probability as the threshold rises. Random forest substantially improves the distribution but remains below the observed curve over much of the upper tail, consistent with its -14.23% pooled WPD bias. Quantile mapping nearly reproduces the pooled observed curve because it directly matches the marginal distribution. This agreement does not imply better hourly timing or tower-specific Weibull skill.

3.8. High-Wind Stability and Tail-Pooling Sensitivity

Figure 12 directly addresses the sparse > 12 m/s regime. Across 20 repeated stratified experiments, raw ERA5 had a mean RMSE of 3.087 m/s. The RF trained only on the > 12 m/s segment reduced mean RMSE to 1.570 m/s, whereas the pooled > 9 m/s RF

reduced it further to 1.461 m/s. The pooled strategy achieved the lower RMSE in 17 of 20 repetitions (85%), with a 95% percentile interval of 1.247–1.610 m/s. Its mean high-wind WPD bias was -3.75% , compared with -4.56% for the separate >12 m/s model and -9.33% for raw ERA5. The tower-block bootstrap gave a mean high-wind RMSE of 1.469 m/s for random forest, compared with 1.503 for LightGBM, 1.525 for XGBoost, 1.595 for CatBoost, and 3.139 m/s for raw ERA5. These results support pooling the sparse upper tail during training while retaining separate >12 m/s evaluation.

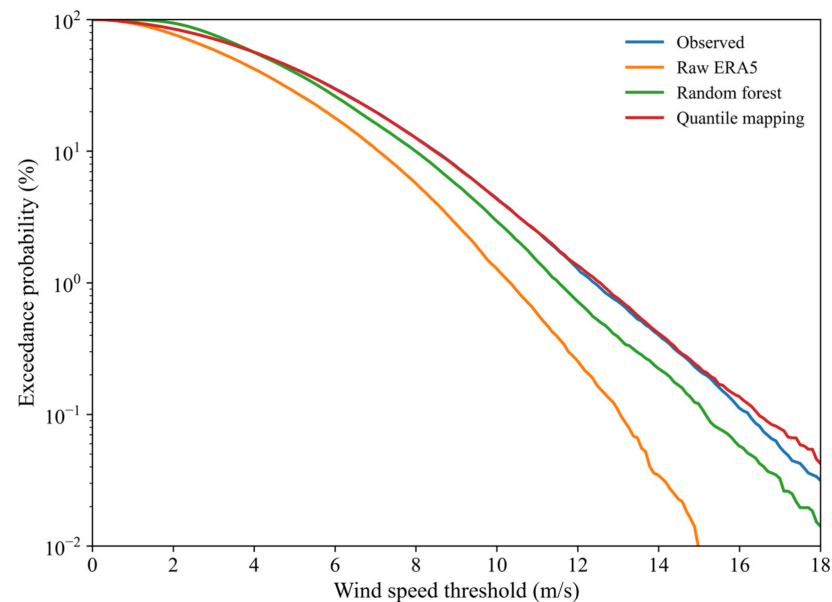


Figure 11. Wind-speed exceedance probability for observations, raw ERA5, random forest, and quantile mapping. The logarithmic y-axis emphasizes the upper tail.

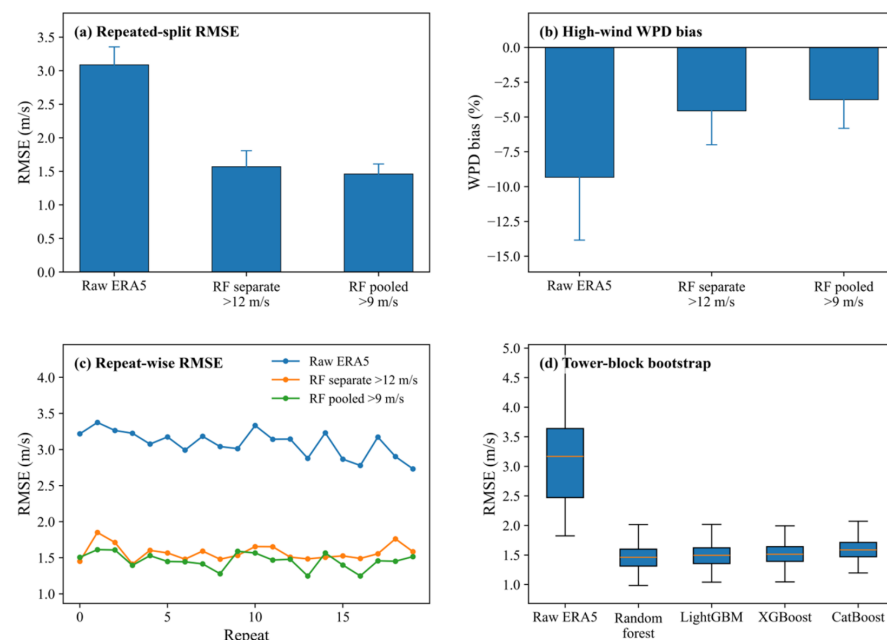


Figure 12. Stability diagnostics for the high-wind regime. (a) Mean RMSE and percentile-based 95% intervals from 20 repeated stratified splits; (b) mean high-wind WPD bias with one standard deviation; (c) repeat-wise RMSE; (d) tower-block bootstrap distributions of high-wind RMSE for raw ERA5 and the four tree-based models.

3.9. Tower-Level Weibull Distribution Assessment

Figure 13 summarizes tower-level two-parameter Weibull diagnostics for all 64 masts. The observed series had mean $k = 1.891$ and $c = 5.394$ m/s. Raw ERA5 reproduced the Weibull shape relatively closely (mean $k = 1.966$) but underestimated the scale parameter (mean $c = 4.338$ m/s). Quantile mapping yielded mean $k = 2.014$ and $c = 5.344$ m/s. Random forest produced mean $k = 2.383$ and $c = 5.428$ m/s; although its fitted shape departed more strongly from the observed Weibull shape, it had the smallest scale-parameter MAE (0.064 m/s), Weibull-derived WPD MAE (22.59 W/m²), and capacity-factor MAE (0.0175), with across-tower correlations of 0.979 and 0.989 for Weibull-derived WPD and capacity factor, respectively.

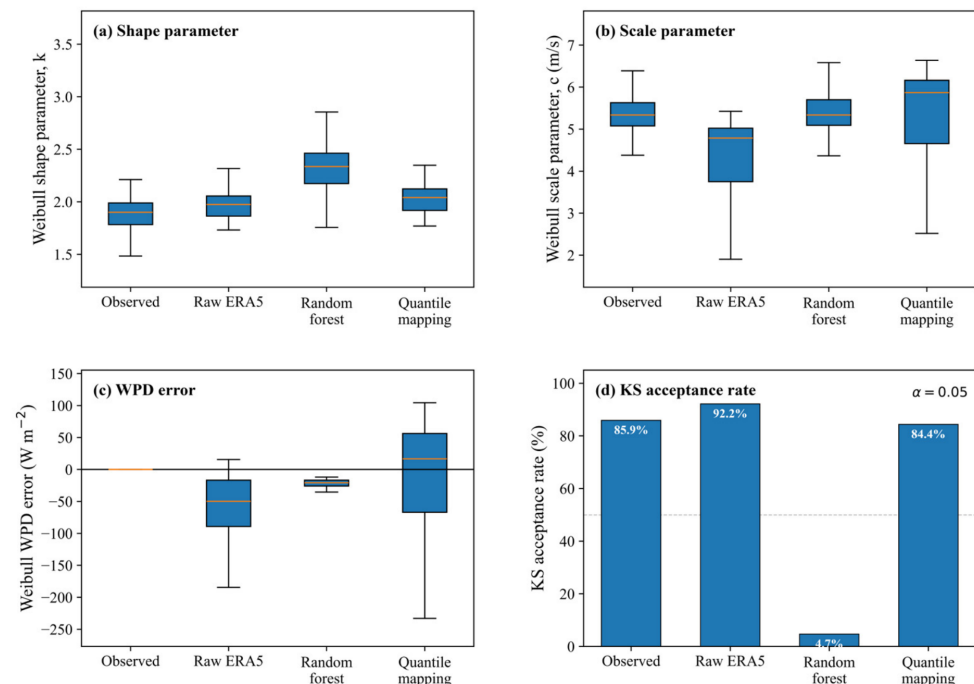


Figure 13. Tower-level two-parameter Weibull assessment for 64 wind masts. (a) Weibull shape parameter k ; (b) Weibull scale parameter c ; (c) Weibull-derived WPD error relative to the observed-wind reference; and (d) KS acceptance rate at $\alpha = 0.05$. Orange horizontal lines in panels (a–c) denote the medians. KS p -values are used as descriptive goodness-of-fit diagnostics because Weibull parameters were estimated from the same tower samples.

At $\alpha = 0.05$, the descriptive one-sample KS acceptance rates were 85.9% (55/64) for observations, 92.2% (59/64) for raw ERA5, 84.4% (54/64) for quantile mapping, and only 4.7% (3/64) for random forest. This result does not imply that raw ERA5 is more accurate overall: a series may resemble a two-parameter Weibull distribution while still having biased scale and energy content. Conversely, the tree-corrected series can improve paired errors and tower-level energy variation while departing from an exact two-parameter Weibull form. The Weibull and KS results are therefore used as complementary distributional diagnostics rather than as the sole criterion for correction quality.

4. Discussion

ERA5 exhibited a persistent low bias across the 64-mast network, and the bias became more negative as observed wind speed increased. This behavior is consistent with recent evidence that global reanalysis performance depends on terrain, height, location, and the metric used [16–20]. Coarse orography and unresolved local flow are plausible contributors, but the present data do not permit attribution to a single physical mechanism. The result

should therefore be interpreted as a regional empirical limitation of ERA5 rather than a universal bias magnitude.

Relative to previous studies, the present results place the evaluated models between simple strong-wind correction and more complex machine-learning approaches. Gandoin and Garza [34] proposed a physically motivated correction for the strong-wind underestimation of ERA5 in offshore conditions, whereas Hallgren et al. [35] applied random forest and neural-network models to predict coastal wind-speed profiles from single-level ERA5 variables and emphasized that improvement in one error metric does not necessarily preserve the full wind-speed distribution. Houndekindo and Ouarda [20] further demonstrated the potential of LSTM and Transformer architectures for hourly ERA5 wind-speed bias correction. The present random forest result should therefore not be interpreted as universal superiority. It indicates that, for the current tabular predictor set, regional sample, and segmented residual target, bagging achieved the lowest held-out paired RMSE, while quantile mapping remained more accurate for the pooled distributional wind-energy indicators. The contribution of this comparison is consequently the controlled within-dataset evaluation and regime-specific validation rather than a claim that one algorithm is optimal for every application.

Random forest provided the most balanced paired hourly correction among the four tree methods. Its advantage over the global RF was modest in aggregate RMSE (1.433 versus 1.455 m/s), but segmentation supports regime-specific behavior and explicit control of the high-wind tail. Pooling the >9 m/s training tail increased the high-wind training sample from 929 to 10,155 records and produced the lower >12 m/s RMSE in 17 of 20 repeated experiments. This result qualifies the assumption that flexible tree models alone can overcome severe sample imbalance: model complexity does not replace adequate tail support, and the regime design and validation strategy remain important.

The terrain analysis provides complementary evidence for the role of topographic complexity. The strong tower-level association between 5-km terrain relief and raw ERA5 RMSE (Pearson $r = 0.872$; Spearman $\rho = 0.800$) indicates that larger unresolved elevation variability is associated with larger site-level reanalysis error in this regional network. However, adding the relief index to the random forest improved RMSE by only 0.71% in the controlled ablation. The two findings are not contradictory: terrain complexity is strongly associated with the magnitude of the original ERA5 error, whereas a single scalar relief descriptor captures only part of the complex local-flow information that could be useful for statistical correction. The association should therefore not be interpreted as causal attribution to terrain alone.

A key implication is that paired time-series accuracy, pooled distributional accuracy, and tower-level energy accuracy are distinct objectives. Random forest minimized paired RMSE and reconstructed diurnal and monthly variation, but its smoothing of the upper tail left WPD and capacity factor low. Pooled quantile mapping nearly restored the overall exceedance curve and aggregate energy indicators, yet it did not improve paired timing and did not preserve tower-level Weibull scale or energy differences. The tower-level Weibull results therefore refine the pooled conclusion: quantile mapping is effective for the aggregate marginal distribution, whereas random forest is more useful for local paired reconstruction and between-tower energy variation. A hybrid approach that combines residual correction with site-aware distributional calibration is a logical next step.

The robustness tests highlight the importance of validation design. Random splitting produced a 36.61% RMSE reduction, whereas temporal extrapolation and spatial tower holdout produced smaller reductions of 14.15% and 14.47%, respectively. Unlike the earlier random-split result alone, both stricter experiments demonstrate positive but more moderate transferability: R increased from 0.666 to 0.689 in temporal extrapolation and

from 0.677 to 0.726 in spatial holdout, while R^2 increased from 0.279 to 0.468 and from 0.352 to 0.526, respectively. The tower-bias ablation further showed that historical tower descriptors were unnecessary for the primary model, reducing site-specific dependence and providing a more defensible basis for spatial transfer.

Several limitations remain. First, although 1162 valid paired records fell in the ERA5 >12 m/s range, only 233 were contained in the main validation subset, so uncertainty in the extreme tail remains appreciable. Second, the 64 masts represent one regional complex-terrain domain, and transfer to other climates and terrain types requires independent validation. Third, air density was estimated from the standard atmosphere at terrain elevation plus 100 m because matched pressure and temperature were unavailable; hourly thermodynamic variability was therefore not represented. Fourth, no coincident, quality-controlled wind-farm SCADA or actual generation series was available for the 64 mast sites. Consequently, the WPD, capacity factor, and equivalent full-load hours are theoretical resource diagnostics derived from wind speed and a generic normalized turbine curve, not direct observations of plant generation. Fifth, the generic power curve does not represent turbine-specific control, availability, wakes, curtailment, or electrical losses and therefore cannot support bankable energy-yield claims. Sixth, the KS p -values are descriptive because Weibull parameters were estimated from the same tower samples, and exact two-parameter Weibull conformity should not be treated as the sole measure of correction quality. Finally, the tree-based correction is empirical and should not be extrapolated beyond the predictor, terrain, and wind-speed ranges represented in training.

5. Conclusions

This study evaluated ERA5 100 m wind speed using 459,630 valid hourly observation–reanalysis pairs from 64 wind masts and developed a terrain-aware, regime-dependent residual-correction framework. Raw ERA5 showed a pronounced low bias in the stratified random validation sample, with $R = 0.669$, $R^2 = 0.307$, $RMSE = 2.261$ m/s, $MAE = 1.664$ m/s, and $ME = -0.904$ m/s. Among the evaluated methods, random forest provided the best paired hourly performance, increasing R to 0.850 and reducing $RMSE$ and MAE to 1.433 and 1.078 m/s, respectively. It improved $RMSE$, MAE , R , and R^2 at all 64 masts.

The additional validation experiments showed that the large improvement obtained under random splitting should not be interpreted as unrestricted transferability. $RMSE$ reductions decreased from 36.61% under random splitting to 14.15% under chronological holdout and 14.47% under spatial tower holdout. Terrain relief was strongly associated with tower-level raw ERA5 error, although its controlled inclusion produced only a modest additional $RMSE$ reduction of 0.71%. Pooling the >9 m/s training tail also improved high-wind stability, outperforming the separately trained >12 m/s model in 17 of 20 repeated experiments.

The preferred correction therefore depends on the intended application. Random forest is more suitable for paired hourly reconstruction and tower-level variation, whereas quantile mapping more closely reproduces the pooled wind-speed distribution and theoretical wind-energy indicators. Because the study did not include coincident wind-farm SCADA data and used a generic normalized turbine power curve, the reported WPD, capacity factor, and equivalent full-load hours should be interpreted as wind-resource diagnostics rather than direct estimates of realized power generation.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/doi/s1>: Table S1: Mast-specific evaluation of raw ERA5 and the primary random forest correction at all 64 wind masts; Table S2: Candidate hyperparameter configurations and internal cross-validation results for the four segmented tree models across the four training

wind-speed regimes; Code S1: Analysis and figure-generation scripts, dependency specification, input-data schema, and execution instructions.

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Data Availability Statement: The complete analysis and figure-generation code, together with the dependency specification, input-data schema, and execution instructions, is provided as Supplementary Code S1. The archive does not contain the restricted wind-mast observations. The wind-mast data are not publicly available because they contain station-level measurement information and are subject to data-use restrictions. Access requests may be directed to the corresponding author and will be considered subject to permission from the data provider.

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Abbreviations

The following abbreviations and symbols are used in this manuscript:

Abbreviation	Definition	
AGL	Above ground level	
CV	Cross-validation	
ECDF	Empirical cumulative distribution function	
ERA5	ECMWF Reanalysis v5	
KS	Kolmogorov–Smirnov	
OBS	Wind-mast observation	
QM	Quantile mapping	
RF	Random forest	
SRTM	Shuttle Radar Topography Mission	
Symbol	Definition	Unit
x_i	ERA5 100 m wind speed for sample i	m/s
y_i	Observed wind speed at 100 m AGL for sample i	m/s
X_i	Predictor vector for sample i	—
e_i	ERA5 residual, defined as $y_i - x_i$	m/s
$\hat{e}_i$	Predicted ERA5 residual	m/s
f_β	Residual model for wind-speed training regime b	—
$\hat{y}_i$	Corrected 100 m wind speed	m/s
$\bar{e}_{\text{train}}$	Mean training residual used in mean-bias correction	m/s
R	Pearson correlation coefficient	—
R^2	Coefficient of determination	—
RMSE	Root mean square error	m/s
MAE	Mean absolute error	m/s
ME	Mean signed error	m/s
nMAE	Mean-normalized mean absolute error	%
z_i	Absolute altitude used for air-density calculation	m
ρ_i	Air density at terrain elevation + 100 m AGL	kg m ^{−3}
v	Wind speed	m/s

WPD _i	Wind power density for sample i	W m ^{−2}
P*(v)	Normalized turbine power at wind speed v	—
CF	Capacity factor	—
EFLH	Equivalent full-load hours	h
k	Weibull shape parameter	—
c	Weibull scale parameter	m/s

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